

Week 2 Progress Report

Project: Log File Analyzer for Intrusion Detection

1. Introduction

In Week 2, the focus shifted from log parsing to implementing detection mechanisms for identifying suspicious activities. The structured data obtained from Week 1 was used to analyze patterns and detect potential security threats.

2. Objectives Achieved

- 1 Implemented brute-force attack detection
- 2 Implemented DoS (Denial-of-Service) detection
- 3 Analyzed frequency of requests using pandas
- 4 Developed modular detection functions

3. Technical Work

Detection logic was implemented using pandas operations such as `value_counts()` to identify abnormal patterns. Brute-force attacks were detected by identifying multiple failed login attempts from a single IP address. DoS attacks were identified by detecting unusually high request rates from specific IPs. Threshold values were defined to classify suspicious behavior. Additionally, modular functions were created in `detector.py` to keep the system scalable and maintainable.

4. Tools and Technologies Used

- 1 Python 3
- 2 pandas for data analysis
- 3 Regex for log pattern matching
- 4 Kali Linux environment

5. Outcome

By the end of Week 2, the system was capable of detecting brute-force and DoS attacks from log data. The detection module successfully identified suspicious IP addresses based on defined thresholds, marking a significant step toward building a functional intrusion detection system.

6. Conclusion

Week 2 enhanced the project by introducing core detection capabilities. The system now not only processes logs but also identifies potential threats, making it more aligned with real-world intrusion detection systems.